Probability Practice Questions

QNO01: What is the probability that the numbers 11, 4, 17, 39, and 23 are drawn in that order from a bin containing 50 balls labeled with the numbers 1, 2,..., 50 if

1. the ball selected is not returned to the bin before the next ball is selected
2. the ball selected is returned to the bin before the next ball is selected?

The probability of drawing the number 11 first is 1/50 (since there is only one ball labeled 11 out of 50). The probability of drawing the number 4 second is 1/49 (since there is only one ball labeled 4 remaining out of 49 after the first ball is drawn). Similarly, the probabilities for drawing the numbers 17, 39, and 23 in the third, fourth, and fifth positions respectively are 1/48, 1/47, and 1/46. To find the overall probability of drawing the numbers in that specific order, we multiply these individual probabilities together: P(11, 4, 17, 39, 23) = (1/50) * (1/49) * (1/48) * (1/47) * (1/46) ≈ 1/254,251,200.	The probability of drawing the number 11 first is still 1/50. The probability of drawing the number 4 second is still 1/50. Similarly, the probabilities for drawing the numbers 17, 39, and 23 in the third, fourth, and fifth positions respectively are still 1/50. To find the overall probability of drawing the numbers in that specific order, we multiply these individual probabilities together: P(11, 4, 17, 39, 23) = (1/50) * (1/50) * (1/50) * (1/50) * (1/50) = 1/3125000000 So, the probability of drawing the numbers 11, 4, 17, 39, and 23 in that order, with replacement, from a bin containing 50 balls is 1 in 3,125,000,000

QNO02: What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?

Divisible by 2: There are 50 positive integers not exceeding 100 that are divisible by 2 (2, 4, 6, ..., 98, 100). Divisible by 5:

There are 20 positive integers not exceeding 100 that are divisible by 5 (5, 10, 15, ..., 95, 100).

However, we need to be careful not to count the numbers that are divisible by both 2 and 5 (i.e., divisible by 10) twice. So, we subtract the number of positive integers divisible by both 2 and 5 (10, 20, 30, ..., 90, 100).

Number divisible by either 2 or 5 = (Number divisible by 2) + (Number divisible by 5) - (Number divisible by both 2 and 5)
= 50 + 20 - 10
= 60

Total number of positive integers not exceeding 100 = 100

Therefore, the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5 is:

P(divisible by 2 or 5) = (Number divisible by either 2 or 5) / (Total number of positive integers not exceeding 100)
= 60 / 100
= 3/5
≈ 0.6

So, the probability is approximately 0.6 or 60%.

QNO03: What is the probability that a player of a lottery wins the prize offered for correctly choosing five (but not six) numbers out of six integers chosen at random from the integers between 1 and 40, inclusive?

otal possible outcomes:
The total number of ways to choose six numbers out of 40 is given by the combination formula: C(40, 6) = 40! / (6! * (40-6)!) = 3,838,380.

Favorable outcomes:
To correctly choose five numbers out of six, we have to consider that there are 40 numbers to choose from and we need to select five of them correctly. The remaining number can be any of the remaining 34 numbers.

The number of ways to choose five numbers correctly out of 40 is given by the combination formula: C(40, 5) = 40! / (5! * (40-5)!) = 658,008.

The number of ways to choose the remaining number incorrectly is given by the combination formula: C(34, 1) = 34! / (1! * (34-1)!) = 34.

Therefore, the total number of favorable outcomes is: 658,008 * 34 = 22,349,352.

Probability of winning:
The probability of winning the lottery prize by correctly choosing five out of six numbers is the ratio of favorable outcomes to total possible outcomes:

P(winning) = favorable outcomes / total possible outcomes = 22,349,352 / 3,838,380 ≈ 0.005815

So, the probability of winning the prize is approximately 0.005815 or 0.5815%

QNO04: What probabilities should we assign to the outcomes H (heads) and T (tails) when a fair coin is flipped? What probabilities should be assigned to these outcomes when the coin is biased so that heads comes up twice as often as tails?

P(H) = 0.5
P(T) = 0.5

However, when the coin is biased so that heads comes up twice as often as tails, the probabilities assigned to the outcomes change. In this case, we need to consider the relative frequencies or ratios of the outcomes.

Let's assign a probability of p to tails (T). Since heads (H) comes up twice as often, its probability will be 2p.

To determine the probabilities, we need to find the values of p and 2p that satisfy the condition that the sum of the probabilities equals 1.

p + 2p = 1
3p = 1
p = 1/3

Therefore, the probabilities assigned to the outcomes when the coin is biased are:

$P(H) = 2p = 2 * (1/3) = 2/3$
$P(T) = p = 1/3$

So, the probability of getting heads (H) is 2/3 or approximately 0.6667, and the probability of getting tails (T) is 1/3 or approximately 0.3333.

QNO05: Find the smallest number of people you need to choose at random so that the probability that at least one of them has a birthday today exceeds 1/2.

For 1 person:
Probability of no matches = (364/365)
Probability of at least one match = 1 - (364/365) = 1/365

For 2 people:
Probability of no matches = (364/365) * (363/365)
Probability of at least one match = 1 - ((364/365) * (363/365))

QNO06: What is the conditional probability that exactly four heads appear when a fair coin is flipped five times, given that the first flip came up tails?

E: Four heads appear when a coin is flipped five times
F: The first flip is a tails
E = {HHHHT, HHHTH, HHTHH, HTHHH, THHHH}; |E| = 5
$|S| = 2^5 = 32$
$p(E) = 5/32$
$p(F) = 1/2$
$p(E \cap F) = 1/32$
since $E \cap F = \{THHHH\}$
$p(E)p(F) = 5/32 * 1/2 = 0.078125$
$p(E)p(F) \neq p(E \cap F)$, so the events are dependent
$p(E|F) = p(E \cap F)/p(F)$
$=1/32 / 1/2 = 0.0625$

QNO07: Suppose that one person in 10,000 people has a rare genetic disease. There is an excellent test for the disease; 99.9% of people with the disease test positive and only 0.02% who do not have the disease test positive.

1. What is the probability that someone who tests positive has the genetic disease?
2. b) What is the probability that someone who tests negative does not have the disease?

First, determine the events: let A be the event that someone has the disease, and B the event that
someone's test is positive. From the problem we know:
p(A) = 1/10,000
p(B|A) = 0.999
p(B|~A) = 0.0002
What is the probability that someone who tests positive has the disease? We want to find the probability
of A given B, p(A|B). Using Bayes theorem we have:

$$p(A|B) = \frac{p(B|A) * p(A)}{p(B)}$$
$$p(B) = p(A) * p(B|A) + p(\sim A) * p(B|\sim A)$$
$$p(B) = (\frac{1}{10000}) * (0.999) + (1 - \frac{1}{10000}) * (0.0002)$$
$$p(B) = 0.00029988$$

Now we can solve for p(A|B)

$$p(A|B) = \frac{0.999 * 0.0001}{0.00029988}$$
$$p(A|B) = 0.333 \textbf{ (4 pts)}$$

What is the probability that someone who tests negative does not have the disease? We want to find the probability of ~A given ~B, p(~A|~B). Using the definition of conditional probability we have:

$$p(\sim A|\sim B) = \frac{p(\sim A \cap \sim B)}{p(\sim B)}$$

Note: $p(\sim B) = p(\sim A \cap \sim B) + p(A \cap \sim B) = (.9999)(.9998) + (.0001)(.001) = .99970012$

Thus, we have $p(\sim A|\sim B) = \frac{p(\sim A \cap \sim B)}{p(\sim B)} = \frac{.9999 \text{x} .9998}{.99970012} = .9999999$ **(3 pts)**